

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE**

**Bachelor of Technology (Mechanical Engineering) SEMESTER - 3 Winter 2025 ( Regular )**

**Course :Bachelor of Technology (Mechanical Engineering) Branch : Engineering and Technology**

**Semester : SEMESTER - 3**

**Subject Code & Name: 24UD1612PC302 - MATERIALS SCIENCE AND METALLURGY**

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**Time : 3 Hours]**

**[Total Marks : 60**

**Instructions to the Students:**

1. Each question carries 12 marks.
  2. Question No. 1 will be compulsory and include objective-type questions.
  3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
  4. Use of non-programmable scientific calculators is allowed.
  5. Assume suitable data wherever necessary and mention it clearly.
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**Q1. Objective type questions. (Compulsory Question) 12**

- 1 Average number of atoms per unit cell of HCP crystal structure are ----- . a) 2 b) 4 c) 6 d) 1
- 2 Grain boundaries are ----- defects. a) 0-D b) 1-D c) 2-D d) 3-D
- 3 The number of favorable slip systems is highest in -----crystal structures. a) BCC b) FCC c) HCP d) None of these
- 4 ----- is the resistance of the material to plastic deformation. a) Toughness b) Stiffness c) Hardness d) None of these
- 5 ----- indenter is used during Vicker's hardness test. a) Spherical b) Conical c) Triangular pyramid d) Square pyramid
- 6 If the atomic sizes of solute and solvent differ by less than ----- %, it is said to have a favourable size factor for solid solution formation. a) 15 b) 50 c) 25 d) None of these
- 7 ----- is the hardest phase in the iron-iron carbide equilibrium diagram. a) Ferrite b) Austenite c) Cementite d) Pearlite
- 8 Martensite is a supersaturated solid solution of carbon in BCC iron having ----- structure. a) FCC b) BCC c) BCT d) HCP
- 9 The main purpose of conventional annealing is/are to: a) Relieve internal stresses b) Reduce hardness c) Refine grain size d) All of these
- 10 The main purpose of hardening treatment is to increase the hardness by austenite to ----- transformation. a) Ferrite b) Pearlite c) Martensite d) Bainite
- 11 During etching of single phase alloys and pure metals, the grain boundary regions corrode ----- than/as the grains. a) Faster b) Slower c) At the same rate d) None of these
- 12 Spark test makes use of the following feature(s) of the spark: a) Carrier lines b) Spurs c) Both 'a' and 'b' d) None of these

**Q2. Solve the following.**

- A) Explain the basic characteristics of crystal structures like average number of atoms per unit cell, coordination number, and atomic packing factor. Give the values of these parameters for FCC and HCP crystal structures. 6

- B) Discuss the crystal defects w.r.t. importance and classification. Explain planar defects in detail using neat sketches. 6
- Q3. Solve the following.
- A) Explain slip mechanism of plastic deformation using schematic diagrams. Compare slip with twinning mechanism. What is Schmid's law? 6
- B) Define Engineering stress, True stress, Engineering strain, and true strain. Derive the formulae for relationship between engineering and true stress – strain. 6
- Q4. Solve Any Two of the following.
- A) Describe Gibb's phase rule. Explain solidification of a pure metal using schematic diagram. Draw cooling curve for pure metals. 6
- B) Give classification and applications of steels on the basis of carbon percentage. Explain the specification of steels as per Indian and American standards with the help of suitable examples. 6
- C) Discuss the transformation of austenite to pearlite and bainite using suitable diagrams. 6
- Q5. Solve Any Two of the following.
- A) Explain the heat treatment of steels w.r.t. definition, purposes, heating media, soaking, and cooling media. 6
- B) Discuss hardening process w.r.t. purpose, and types. What is tempering process? Explain its purposes, process details and types. 6
- C) Elaborate gas carburizing and gas nitriding processes w.r.t. mechanism, process details, and schematic diagrams. 6
- Q6. Solve Any Two of the following.
- A) Define metallography and compare microscopy with macroscopy w.r.t. definition and purposes. Give a flow chart of the specimen preparation for microscopy. 6
- B) Explain the principle of working of optical metallurgical microscope using schematic diagram. Define the magnifying power, numerical aperture, and resolution of the objective lens. 6
- C) Discuss the spark test w.r.t. purpose, mechanism of generation of spark, and the spark patterns for low carbon and high carbon steel. 6

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